

Combating Dirty Data using Data Virtualization

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Abstract—Data is the new Gold of the modern age. Organizational decisions, artificial intelligence, machine learning, time critical processes, analytics, etc. are driven by highly profiled data. Data gives an organization the ability to act in a corresponding situation and hence, keep up with the market. Financial returns are directly associated with the quality of data being used. Today's world has experienced a data explosion with the advent of data sources like Hadoop, data lakes, clusters and many more. Even more, the day to day devices that we use collect data to present meaningful statistics to us. There is an exponential increase in data volumes which are characterized by heterogeneous data formats and disparate data sources which in turn is complicating the task of providing uniform and profiled data to the business users for analysis. Warehousing is also not able to cope up with the on-demand real-time data access. Holistic view of the entire data is extremely difficult to achieve and takes a considerable amount of time which is slowing down an organization's decision making capabilities. The complexity of the current systems has increased due to the interconnected nature of widely distributed systems which share data. Dirty data i.e. data which is inaccurate, inconsistent, unclean or incomplete is spread all across. As all the data is unrequired garbage data, there is no point in developing your business intelligence over this data. Data Clarity used along with Data Virtualization provides an exceptional solution to these problems by handling the complex data workloads with respect to heterogeneous data and disparate data sources and provides a holistic view of the data to the business organization.

Keywords—Data Virtualization, Data Intelligence, Federated Query Engine, Data Clarity, Active clusters, Data life-cycle management, metadata management, Extensible Style sheet Language Transformations, disparate data sources, Extraction, Transformation and Loading.

I. INTRODUCTION

Data has now become the DNA of any entity in the modern age. Before the computer age, this data was kept in the form of manuscripts, literature works, etc. The due to the revolution in the media industry, the data represented changed its form into newspapers, books, articles, etc. With the introduction of computer age, this data is now stored in various human readable or non-readable formats. Audio and video gave birth to a very new data segment. During the course of time, the data storage options also changed from manuscripts to highly efficient memory chips. Prior to the industrial revolution, humans still use to gather this data and take decisions based on the historical data. We still do the same but with highly efficient tools. During this course of development, the complexity to handle this data has also increased and is now creating data bottlenecks in various areas. In the current age of data revolution, we have developed numerous data sources like Hadoop, Big Query,

SAP HANA, Data Lakes, Information Clusters, and many others. Data is now represented in various forms like structured, semi-structured and unstructured formats. The complexity of the current systems has increased due to the interconnected nature of widely distributed systems which share data. This development resulted in data explosion and all kinds of data started flowing into the system.

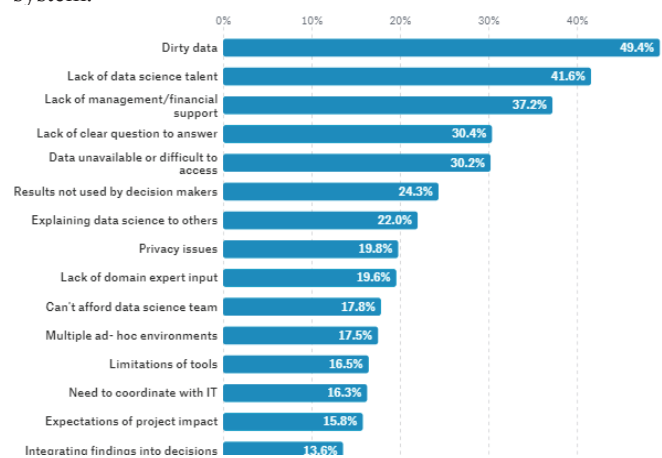


Fig. 1. Data Science Problems.

“Fig 1” showing Dirty Data i.e. the data that is irrelevant, incomplete, absurd, insignificant or erroneous also started flowing into our system. Kaggle conducted an industry survey in 2017, which had 16000+ responses from the people who deal with such data on a day to day basis, and concluded that 49.4% problems in today's industry is a dirty data problems. This has choked the growth all the data dependent organizations. All the organizations are losing revenues and these accounts into \$4 Trillion every year.

II. PROBLEM DEFINITION

A. Increased Data Formats

The increase in the number of structured, unstructured, and semi-structured data formats like RDBMS, JSON, XML, Data Grids, objects, files, etc. has increased the complexity of the query processing system as every time there is a syntax change for different systems and some do not support querying [4]. About 42% of the data scientists feel that it is difficult to extract data from different sources.

B. Longer Data Processing Cycles

Due to garbage data, the data processing cycles have increased and consume 55%-60% of the time of the entire data life-cycle management.

C. Reduced System Availability

The data is not readily available for processing as there are multiple systems to be connected to as the data is scattered in them. The operational efficiency is also being reduced.

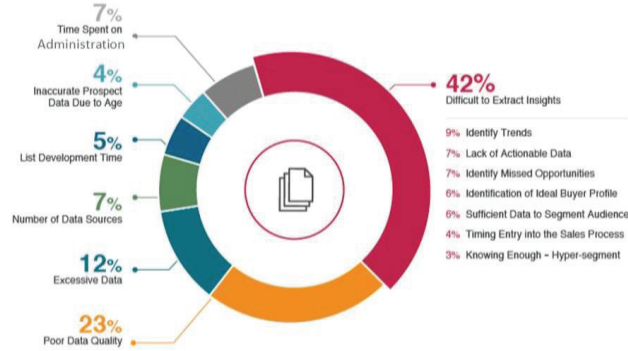


Fig. 2. Dirty Data Segmentation.

D. Industrial problems

The organization's demand for profiled analytical data is exploding and traditional ETL and data warehousing is not keeping pace.

E. Growing data bottlenecks

"Fig 2" showing that, Data bottlenecks due to poor quality of data accounts for 23% of the data problems and are paralyzing the organization's decision making capabilities due to which they are losing revenues and market share [6]. These factors directly impact the Time to Market and are highly correlated with the revenues.

III. DEFINITIONS

A. Data Intelligence

The detailed analysis of the data in a way such that, organizations can utilize the hidden truths in the data for the expansion of their service markets or areas of investments is known as data intelligence [1]. Here the data is curated in such a way that it can be used to prove cases or predicts facts which then can be used for proactive decision making.

B. Data Clarity

Data which has some significance is known as information. The process which is used to transform the raw data into information, by filtering the various aspects in it is known as data cleansing or Data Clarity [6].

C. Data Virtualization

It is a modern data integration approach which abstracts the underlying data complexity, intricate data transformations and provides a holistic view of the entire data system to your business applications [1]. It involves a virtualization layer which handles the complex data federation and provides an ease for the applications to find a single source of true data.

D. Data Life Cycle Management

The process of catering the data right from the stage where it is originated till it is utilized and becomes obsolete is called as Data Life Cycle Management. It involves

various procedures that take place during discovery and capturing, transforming, loading, analysis and then purging phases in the cycle [5].

IV. PROPOSED WORK

"Fig 3" showing, Data Clarity integrated with Data Virtualization can help us in solving the dirty data problem by creating a virtual data layer which carries out the task of cleansing the data, transforming it into the desired format and then combining it with the disparate data sources to create a holistic view of the entire data system.

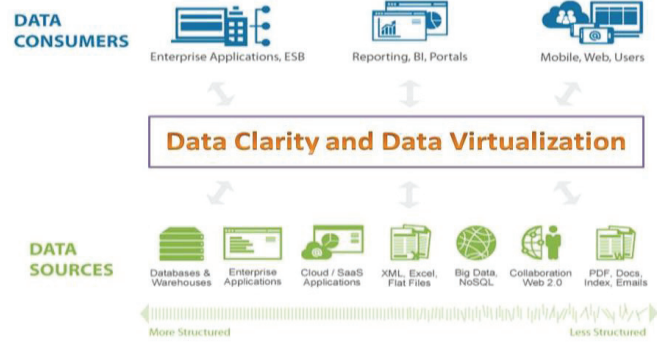


Fig. 3. Data Clarity and Data Virtualization.

Complex data cleansing processes like data batches, clustering of data, de-duplication in data using high efficiency algorithms, can be easily achieved in Data Clarity. When integrated with Data Virtualization, all this data can be brought for other complex transformations like A2A (Any to Any) or XSLT (Extensible Style sheet Language Transformations) so that the archived data becomes easy to combine with other data disparate sources. This holistic view of the system can then be exposed to the outside world of applications without moving any kind of data from its data sources.

"Fig 4" - Data Virtualization consists of various components which are categorized into 3 core environments:



Fig. 4. Data Virtualization - Architecture.

A. Development Environment

In this environment, the entire development of the virtual layer takes place. This section designs the skeleton of the virtual layer which is exposed to the outside world. It consists of 4 main components:

- 1) *Adapters*: These are the connectors that are mainly used to connect to the raw databases containing dirty data.
- 2) *Studio*: This is the agile design time environment where the developers can design their data services, carry out transformations in data and manage the resources.
- 3) *Discovery*: This module helps in establishing complex and hidden relationships between multiple data sources. Going beyond simple data profiling its helps the user to create dynamic entity-relationship models.
- 4) *Business Directory*: This directory keeps a track of the curated data services and helps organizations in the agile decision making.

B. Runtime Environment

In this environment, the runtime aspects of the virtual layer are determined. Performance of the entire data system is determined by this module of the entire system. It consists of 3 main components:

- 1) *Federated Query Engine*: All the query performance is handled by this component. Its prepares a query execution plan to distributed data loads across different sources and achieve optimal query performance..
- 2) *Optimization Engine*: This engine evaluates the data workload dynamically and chooses the best-fit combination of various optimization methods and schema joining algorithms.
- 3) *Metadata Management*: This maintains the entire metadata repository that helps in managing relevant services in the entire development span. It also stores information related to the operations being performed on the data services.

C. Management Environment

In this environment, the entire management of the built data service is carried out. It consists of 2 main components:

- 1) *Deployment manager and Active Clusters*: Deployment managers help in quick migration of data services and also serves in the automation of the data environment. Active clusters are meant for load sharing between the multiple clusters of the system. They dynamically analyze the data workloads and then distribute them accordingly on the multiple clusters.
- 2) *Monitor and Manager*: These are generally used to monitor the health of the system and manages different data services. Administrative tasks are also carried out by these modules

Let us see how the solution fits under the entire Data Life Cycle Management Process.

V. METHODOLOGY

Data Virtualization adopts the following data life-cycle procedure for handling dirty data as shown in “Fig 5”:

A. Data Source Connections

Using the built-in set of connection adapters, any relational or semi-structured or unstructured data sources can be connected. The data sources can also be file systems.

If a corresponding adapter is not present, a quick custom connection can be created using the generic JDBC settings.

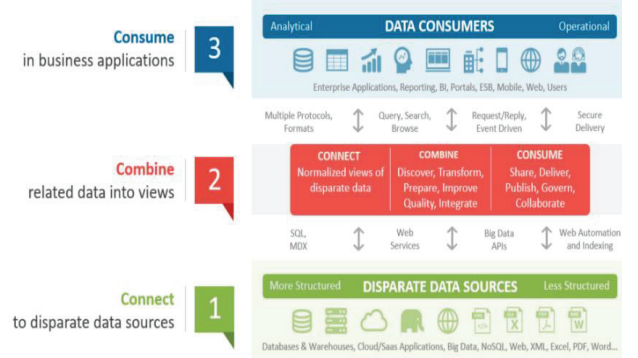


Fig. 5. Data Service Development.

B. Complex Transformations

Data Virtualization automatically recognizes the underlying data source and corresponding data formats. It is well equipped with the data service modeling environment where the users can carry out complex data transformations. Transformations as shown in “Fig 6” can be carried out on XML, XSLT, SQL, XQuery and Java languages.

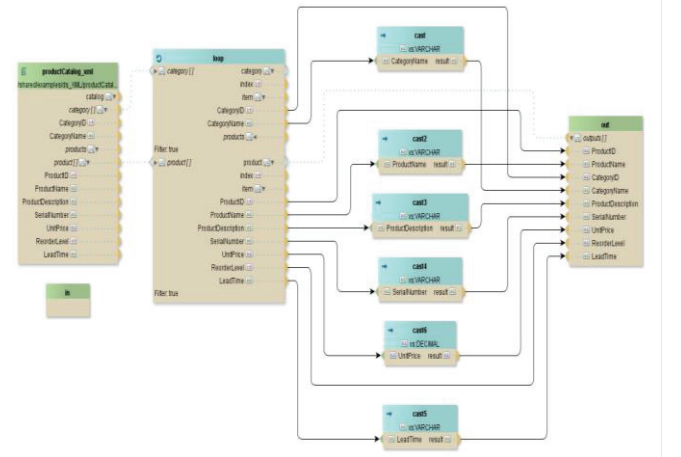


Fig. 6. Complex XSLT Data Transformation.

Its graphical design time environment with efficient code editors provides a workspace where queries can be created and tested. The input section in case of complex transformations consists of the actual raw data source. The editor also allows the developer to create an output format which completely depends on the choice of the developer. Activities can be declared in between the input and the output. Activities are nothing but the actual processes performed on the data. Multiple operations can be carried out on the various attributes of the data. In this way, dirty data can be curated using a low code approach.

C. Data Service Generation

Data from various data sources can be combined together to produce a single unified data service. Data Virtualization provides a drag and drop capability to design this service.

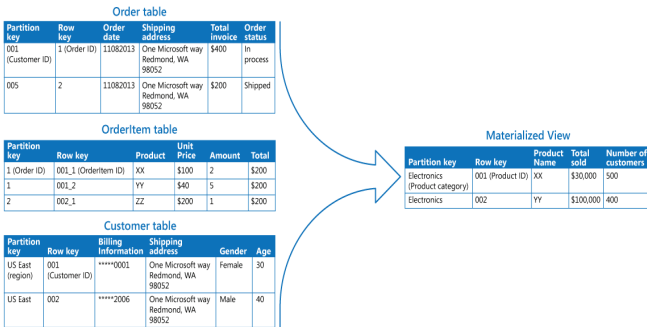


Fig. 7. Data Views.

Development Studio offers a rich set of transformations which are easy-to-use and you can use to transform any source data service structure to any target data service structure. Data from multiple sources can be joined using the primary key/ foreign key relationship. Reuse of the existing services to create dependent services is also possible. Due to this, a holistic view of the entire data system is generated in moments as shown in “Fig 7”.

D. Data Lineage

Data lineage tracks the data from the start of the data life cycle till it is completely curated. In this entire process of data profiling, the source data goes through numerous transformations right from the place of origination. Using the Lineage patterns in “Fig 8”, all the data can be tracked and the corresponding transformations occurring on that data can also be viewed.

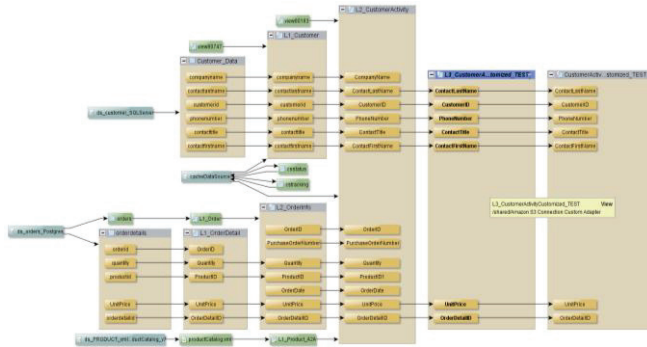


Fig. 8. Data Lineage.

If a problem occurs at any instance of time, it can be tracked using the lineage and the corresponding actions can be taken. This data lineage can be easily tracked in Data Virtualization. The layer which is responsible for all the transformations maintains a pattern of all the data and also keeps a track of its origin. This gives the ability to the developer to undo or redo any changes in the entire lifecycle. Errors can be tracked at any point in the cycle and its root can be easily traced.

E. Data Analytics

The newly created data service can now be exposed to the outside world of analytical applications for consumption as shown in “Fig 9”. Data service developed can be exposed in the form of a database or as a web service as well. Data

Virtualization has a built in connection module that can be used to expose the data service as a database to the outside world. The JDBC connections to the virtualization layer are exposed for the consumption of the data views.

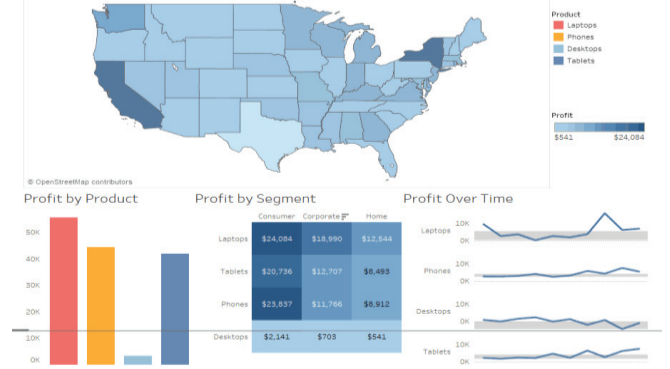


Fig. 9. Data Dashboarding.

Data Virtualization also has a web layer which can be utilized to expose the data as a web service or an API. The development module can be used to wrap the data service with a web layer assigning the protocols and the format of data access. All the major formats of data access through web, like XML, JSON, SOAP, etc. are supported.

VI. RESULTS ANALYSIS

Data Virtualization and Data Clarity helps the developers at various stages of the Data Life Cycle Management. It simplifies various data transformations and reduces the data extraction time to a great extent and helps in combating dirty data. Developing and managing eagle eyed view of the data becomes easier.

TABLE I: Data Virtualization vs. ETL

Sr. No.	Comparison Pointers	Data Virtualization	ETL Process
1.	Data Movement	Not Required	Required
2.	Data Connectors	Available	Not Available
3.	Clustered Deployment	Available	Not Available
4.	Dynamic Query Federation	Available	Not Available
5.	Scalability and Availability	Available	Constant

^a Efficiency in Data Virtualization

A. No Movement in data

The entire data pipeline can be curated without moving a single bit of data. This directly helps by saving the time spent in data loads. Crashes in the target system can also be avoided.

B. Data Connectivity

Due to built-in data connection adapters and the ability to create custom connections, data can be made easily accessible.

C. Complex Transformations

The time and resources spent on handling complex transformations is also drastically reduced. Data transformations to and from any formats can be done. Also, these transformations can be reused for other data sources which give the reusability factor in development.

D. High Performance

Due to the availability of Federated Query Engine and Optimization engine, higher performances can be achieved in less time without complex development cycles.

E. Exposing data

Making the curated data available to the outside world becomes extremely easy. Data can be exposed in the form of API's or as a database itself.

VII. PERFORMANCE ANALYSIS

Performance indicators include various factors like system up-time, scalability, availability and upgrades along with data life cycle and without disturbing it. Let us see how the solution achieves these pointers.

A. Scalability

As data virtualization uses clusters as a mode of deployment, the entire system is scalable horizontally. The number of nodes to be deployed can be decided dynamically with the workloads [3]. Also, the solution can be scaled vertically by deciding on the capabilities in terms of resources each cluster uses.

B. Availability and upgrades

During upgrades, the clusters are managed in a Round Robin fashion which ensures the high availability of the system.

C. Data Extraction

Due to the availability of the connection adapters and the ability to create custom adapters, nearly 7% of the data life cycle time can be saved. The built in transformations easily convert the data into desired table formats without any user intervention.

D. Dirty data handling

Complex Data Transformations can be easily carried out in Data Virtual layer using the low code approach. It saves nearly 5% of the overall time required.

E. Extract data insights

The Discovery module helps in analysing the data and getting hidden relationships in them in a few clicks. This helps in getting deeper insights and in turn tackle about 7% of the missing opportunities.

F. Data relocation

The complete time of data relocation is saved as there is no movement in the data. All the operations are done in the virtualization layer on the fly [2].

VIII. CONCLUSIONS

In this paper, we have represented that Data Virtualization is data integration technology that lets you integrate with big data volumes spread across disparate data sources whether relational or web sources, with breakthrough speed and low code approach. It also helps you in building and managing data services that are virtualized and can access, handle, and deliver profiled data your business contributing towards accelerating revenue streams, reducing operational costs, combating risks, along with improving data compliance, and much more. Also, it has provided a new way to handle intricate data transformations and batch data processing without human intervention.

This research has shown that we can reduce the data life cycle timeline by almost 40% and the problems pertaining to dirty data can be resolved with greater efficiency. While a number of different methods are in competition with each other like ETL, Data Virtualization appears to have the tremendous potential for the near future.

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